

LOHIT

Hisar, Haryana, India | Open to relocation

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SUMMARY

Backend engineer, available immediately. Built Forge, an AI incident-investigation backend with async job processing, pgvector search, and multi-tenant isolation; built dex, a Go durable-execution engine verified by deterministic simulation; and shipped an internal CRM to a team at Smartworks. Depth in Node.js/Fastify, PostgreSQL, and Redis/BullMQ, with working knowledge of Go, Docker, and CI/CD.

EXPERIENCE

Smartworks — Gurugram | Software Development Intern

Jun 2025 – Sep 2025

- Shipped an internal CRM by self-hosting and customizing Twenty (open-source CRM), taking it from a local evaluation setup to a team-facing internal product.
- Implemented push notifications via FCM for WorkCtrl, a React Native app live on the App Store and Play Store.
- Debugged KPI-counting and ticket-assignment defects in the internal ticketing system, tracing discrepancies through the MongoDB aggregation pipeline.
- Built Elasticsearch-backed search across snag and user records, and integrated Formbricks as a feedback widget across the WorkCtrl app and website.
- Built a face-recognition kiosk using OpenCV with supporting automation scripts.

PROJECTS

dex — Durable Execution Engine | [GitHub](#)

Go, PostgreSQL, deterministic simulation testing

- Built a Temporal-style durable execution engine in Go where workflows resume from exactly where they stopped after a crash, by replaying an append-only event log rather than persisting runtime state.
- Designed a single-goroutine deterministic simulator that drives every node, the clock, and the scheduler off one seeded PRNG, enabling reproducible fault injection and shrinking of failing runs — one real triage reduced 44 injected faults to 2.
- Implemented shard leases with fencing, transfer/timer queues with ack levels, and a determinism linter that flags non-replayable calls (e.g. `time.Now`) in workflow code.
- Verified crash recovery against a real process killed mid-execution on Postgres; shipped `dexctl` (CLI) and an embedded web dashboard for inspecting event histories and live workflow state.

Forge — AI Incident-Investigation Backend | [Live Demo](#) | [GitHub](#)

Node.js, Fastify, PostgreSQL/pgvector, Redis/BullMQ, Groq, Gemini, Railway

- Sole developer of a backend that ingests alerts from monitoring tools (Grafana, PagerDuty, Sentry, CloudWatch, Opsgenie) and runs autonomous LLM-driven root-cause investigations.
- Architected a single-process Fastify 5 service running the API alongside three background workers (BullMQ/Redis) on free-tier hosting, backed by Postgres + pgvector.
- Designed a provider-agnostic LLM seam (Groq for generation, Gemini for embeddings) with two-tier model routing and quota-aware rate limiting to operate within free-tier caps.
- Built a resilient ingestion pipeline that normalizes heterogeneous webhook payloads, fingerprints incidents, and prevents duplicate/racing incidents via atomic Postgres upserts, with field redaction, at-rest encryption, and RBAC-based multi-tenant isolation.

A4 Satta King — Real-Time Results Platform

Node.js, Express.js, SQLite, Socket.io, JWT

- Built a full-stack results platform — public site plus a JWT-authenticated admin panel — pushing live updates to all connected clients via Socket.io, backed by a SQLite schema and REST API with rate limiting.

TECHNICAL SKILLS

Proficient: JavaScript (ES6+), TypeScript, Node.js, Fastify, Express.js, REST APIs, PostgreSQL, MongoDB, Redis, BullMQ, Git/GitHub, Linux

Familiar: Go, Docker, GitHub Actions, CI/CD, Python, Java, C/C++, SQL, React.js, Next.js, React Native, Elasticsearch, AWS (EC2, S3, IAM), GraphQL, Drizzle ORM, Tailwind CSS

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP, Embedded Systems

EDUCATION

Guru Jambheshwar University of Science & Technology — Hisar, Haryana | B.Tech, *Graduated May 2026*
Electronics & Communication Engineering

CCSHAU Campus School — Hisar | Senior Secondary Education — 80%

2019 – 2021

LEADERSHIP & VOLUNTEER

- Smart India Hackathon (SIH) Qualifier — Qualified at college level, leading development of an alumni association platform for a government engineering college, focusing on scalable architecture and admin analytics.
- Training & Placement Cell Website Manager — Managed the official T&P Cell website of Guru Jambheshwar University, maintaining functionality and updates for recruiters and students.